

Quantifying Explanation Faithfulness and Localization in CNNs vs. Vision Transformers for Skin Lesion Classification

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Abstract

Deep learning models now match dermatologist-level accuracy at skin-lesion classification, yet clinical adoption remains low because clinicians cannot verify which visual features drive model predictions. This work compares EfficientNet-B0 (CNN) and ViT-B/16 (Vision Transformer) on HAM10000 across three axes: classification accuracy, explanation faithfulness, whether hiding or restoring a heatmap’s most important pixels changes the prediction, and explanation localization, whether the heatmap lands on the lesion an expert outlined. All 1,002 test images are scored on a lesion-grouped split, so no lesion appears in both training and test, and all three axes are measured on the same tuned checkpoints. Our central finding is that accuracy and explanation quality favor opposite architectures. ViT classifies better, reaching 0.748 test macro-F1, the F1 score averaged equally over the seven classes, against 0.655, yet EfficientNet’s Grad-CAM explains better on every measure of explanation quality we apply. It is far more faithful, its deletion-curve area is 0.284 against 0.528 for randomly ordered pixels, where ViT’s rollout scores 0.661 against its own 0.791, and it localizes far more tightly, with region overlap 0.466 against 0.401 and its most important pixel inside the lesion 98% of the time against 65%. Insertion agrees: adding back the highest-ranked pixels first recovers the prediction with area 0.736 against 0.283 for random order, where ViT reaches 0.582 against 0.387. Two training choices proved architecture-dependent: linear probing then fine-tuning beat every partially-unfrozen alternative on both models, while data augmentation, not dropout, is what narrows EfficientNet’s train-validation gap and the same augmentation reduces ViT’s validation performance at every setting tested. Together these results caution that a model chosen on accuracy alone may be the weaker one to show a clinician.

1. Introduction

Skin cancer is common and early detection saves lives, but there are far too few dermatologists to screen everyone. A computer program can now judge from a photograph, about as accurately as a dermatologist, whether a skin spot is harmless or dangerous, which raises the prospect of a general doctor, or even a patient, flagging a dangerous lesion early. The obstacle is trust: the program gives no reason for its answer, so a doctor cannot tell whether it looked at the spot itself or at something irrelevant beside it, such as a ruler mark or a hair. A doctor will only act on a model that can show which part of the image led to its decision. We ask whether the explanations these programs offer can be trusted.

Since Esteva *et al.* [1] matched dermatologists at skin-cancer classification, the standard recipe fine-tunes a pre-trained model on a dataset such as HAM10000 [2] and reports accuracy. Recent work compares the two dominant architectures on this task, convolutional networks like EfficientNet [3] against Vision Transformers like ViT [4]; Kuntal and Bhat [5] do so on HAM10000 but on accuracy alone. To explain the models, practitioners overlay a heatmap, Grad-CAM [6] for CNNs and attention rollout [7] for Transformers, and judge it by eye or by its overlap with an expert’s lesion outline; the closest work to ours, Koza-chok and Seregin [8], does exactly this on roughly 180 cases as product validation. Two limits follow. Accuracy says nothing about *why* a model decides, and overlap measures only *where* a heatmap falls, not whether those pixels actually drive the prediction: a map can sit on the lesion yet be unfaithful. Faithfulness tests such as RISE [9], which removes or adds pixels in order of a heatmap’s importance scores and measures how much the prediction moves, and perturbation curves [10] address the second question, but they have not been paired with localization in a matched CNN-versus-ViT comparison.

We close that gap. We train two models of very different internal design, EfficientNet-B0 and ViT-B/16, on the same seven lesion types, produce a heat map for each pre-

diction, and test every map two ways: whether hiding or restoring the highlighted regions changes the model’s answer, and whether those regions land on the lesion an expert outlined. This measures both the honesty and the accuracy of each model’s explanations, so that a doctor could rely on an AI diagnosis and understand the reason behind it.

We use HAM10000 [2], 10,015 dermoscopic images labeled by histopathology or expert consensus into seven lesion types. Following the datasheet framework [11], we highlight the aspects that shape our design. **Composition:** the classes are severely imbalanced, from melanocytic nevus at 67% to dermatofibroma at 1%, which is why we weight the loss and report per-class metrics rather than raw accuracy. Each image also ships with a binary lesion segmentation mask, an expert-drawn outline that marks every pixel belonging to the lesion, the region a clinician reads to make the diagnosis, and separates it from surrounding healthy skin. These masks are the ground truth for our localization tests, which ask whether a model’s heat map falls on the same pixels an expert would, and they are the reason we chose this dataset. The 10,015 images cover only 7,470 unique lesions, since some lesions are photographed more than once, so a random split would leak copies of a lesion across train and test; we use a lesion-grouped stratified 80/10/10 split. **Collection process:** the images are patient health data, so we use only the public de-identified release. We resize every image to 224×224 RGB.

2. Approach

We compare two architectures with opposite inductive biases on the same task. EfficientNet-B0 [3] is a convolutional network whose receptive fields grow with depth, so it reasons from local texture outward; ViT-B/16 [4] splits the image into patches and lets every patch attend to every other from the first layer, so it can relate distant regions immediately. Contrasting a locality-biased model against a global-attention model is what lets us ask whether architecture shapes not just accuracy but the honesty and placement of a model’s explanation. We start from the public ImageNet checkpoints on Hugging Face, `google/efficientnet-b0` and `google/vit-base-patch16-224`, and replace each classifier head with a fresh seven-way layer for our lesion classes, keeping the pretrained backbone. All evaluation and training code is our own and public.¹

¹Code: <https://github.com/SebastianGonzalez2000/dermai-explainability>

Base models:

<https://huggingface.co/google/efficientnet-b0>,
<https://huggingface.co/google/vit-base-patch16-224>

Fine-tuned models: https://huggingface.co/xyxing/dermai_efficientnet-b0_trained,
https://huggingface.co/xyxing/dermai_vit_trained

We explain each model with the method standard for its architecture. For the CNN we use Grad-CAM [6], which weights the final convolutional feature maps by the gradient of the predicted class to produce a heat map. For the ViT we use attention rollout [7], which multiplies the per-layer attention matrices to trace how much each patch contributes to the final class token. Abnar and Zuidema [7] combine each matrix with the identity matrix before multiplying, since a transformer layer also passes each token forward through a residual connection that raw attention weights ignore. Both methods yield a heat map over the image that we then test.

Explanation quality has two components. Faithfulness asks whether the highlighted pixels actually drive the prediction: following RISE [9], the deletion test removes pixels from most to least important and records how fast the predicted probability falls, while the insertion test adds them back into a blank canvas and records how fast it rises. Localization asks whether the heat map lands on the lesion. We first turn the heat map into a highlighted region by keeping its most important pixels, then score that region against the expert mask two ways. Intersection-over-union is the area the region and the mask share divided by the area they cover together, so 1 means perfect overlap and 0 means none. The pointing game checks the single most important pixel, and counts a hit when it lands inside the mask. Pairing both components in a matched CNN-versus-ViT comparison, rather than reporting accuracy or overlap alone, is the new element of our approach.

Two properties of this setup drive design choices we make up front. ViT-B/16 was pretrained on far more data than EfficientNet-B0 needs to converge, so on a set as small as HAM10000 we expect it to be less stable and give it a smaller Stage 2 learning rate. And a Hugging Face checkpoint’s defaults are tuned for its original task, not ours: the ViT checkpoint ships with `hidden_dropout_prob` and `attention_probs_dropout_prob` at 0.0, so we enable dropout for fine-tuning rather than train on roughly 8,000 images with no internal regularization. A checkpoint’s own defaults matter as much as the schedule layered on top.

We fine-tune in two stages to protect the pretrained features, the linear probing then fine-tuning schedule of Kumar *et al.* [12], hereafter LP-FT. Stage 1 freezes the backbone and trains only the new head; stage 2 unfreezes the backbone and trains end to end at a much smaller learning rate. Training a randomly initialized head against a frozen backbone first matters because the large early gradients of an untrained head otherwise distort the pretrained features they backpropagate through [12]: a head initialized at random produces large gradients on the very first batches, which flow back through an already-good backbone and destructively update it for the sake of a head that has not yet learned

anything. This concern goes back to the gradual unfreezing of Howard and Ruder [13], and Tomihari and Sato [14] trace it to the growth of the head norm during probing. Probing first drives the head to a reasonable solution while the backbone cannot move, so that when the backbone is released the gradients reaching it are already small and informative. Davila *et al.* [16] find the schedule improves over half of the cases they test across five medical imaging modalities, dermoscopy among them. Both models use AdamW [17] and a class-weighted cross-entropy loss to counter the imbalance; learning rates, epoch counts, and regularization settings are selected empirically in Section 3.1.

We did not assume LP-FT was the right schedule. Its alternative is surgical fine-tuning, in which only part of the backbone is ever unfrozen. Lee *et al.* [15] show that tuning a well-chosen subset of layers can match or beat tuning the whole network, and that which subset wins is determined by the kind of distribution shift between pretraining and the target task, with input-level corruptions favoring the earliest layers and label-level shifts the latest. Section 3.1.1 tests restricted unfreezing against full unfreezing on both architectures and finds no configuration that beats LP-FT, so we keep it. That is the outcome the method predicts here: surgical fine-tuning targets a shift localized to one end of the network, whereas moving from ImageNet objects to dermoscopic lesions changes both the low-level image statistics and the label semantics at once, leaving no single block range that isolates the shift.

3. Experiments and Results

We measure success on the three axes set out in Section 2: classification quality, since a model that cannot classify has nothing worth explaining, and explanation quality, faithfulness by deletion and insertion, and localization by IoU and the pointing game.

3.1. Model Selection

3.1.1 Fine-Tuning Strategy Selection

To determine the optimal fine-tuning strategy we compare two schedules across both architectures. **Strategy 1** is LP-FT [12]: freeze the backbone in Phase 1 and vary the Phase 2 unfreeze depth (last {2, 4, 6, all} layers, 10 epochs/setting), a direct test against surgical fine-tuning [15]. **Strategy 2** fixes Phase 2 to a full unfreeze and instead varies the Phase 1 warmup unfreeze depth (last {2, 4, 6} layers), testing whether letting backbone layers adapt before the head is trained helps over a fully frozen warmup. In both, the reinitialized 7-class head is always trainable.

Table 1 summarizes the result (full depth sweep in Appendix A.1). Given HAM10000’s class imbalance we report macro-F1, the F1 score averaged equally over the seven classes so the majority class does not dominate, and bal-

Strategy	EfficientNet		ViT	
	Train-mF1	Val-mF1	Train-mF1	Val-mF1
1 (full unfreeze)	0.9880	0.6874	1.0000	0.7707
2 (best depth)	0.9986	0.6866	1.0000	0.7577

Table 1: Best result per strategy; full depth sweep in Appendix Table 8.

anced accuracy, the recall averaged equally over the seven classes, rather than raw accuracy, throughout. For both architectures, Strategy 1 with a full Phase-2 unfreeze achieves the best validation macro-F1, and within Strategy 1 validation performance rises monotonically with unfreeze depth (0.5631 up to 0.6874 on EfficientNet, 0.6956 up to 0.7707 on ViT), so no restricted depth beats full unfreezing and surgical fine-tuning brings no gain here. Strategy 2 instead shows near-saturated training performance across all Phase-1 unfreeze depths, with validation performance flat and below Strategy 1’s best in every case, consistent with LP-FT’s prediction that unfreezing during warmup lets an untrained head’s gradients reach the pretrained features. Both strategies show a substantial train-validation gap, motivating the overfitting mitigation study below. We retain Strategy 1 (LP-FT), frozen Phase 1 and fully unfrozen Phase 2, as our baseline going forward.

3.1.2 Overfitting Mitigation

To mitigate the overfitting observed in the baseline depth sweep, we test three regularization techniques: dropout (applied to hidden activations for both architectures), drop_connect_rate (stochastic-depth-style regularization applied only to EfficientNet’s MBConv blocks, as this mechanism is specific to convolutional residual architectures and has no ViT equivalent), and train-time data augmentation on the model architecture selected above. We report both average and best validation F1 throughout, as each captures a different property: average F1 reflects a configuration’s typical, more reproducible performance and is less sensitive to a single lucky epoch, while best F1 reflects the value actually captured by our checkpointing criterion and used at test time. Table 2 summarizes their individual effects relative to each architecture’s untuned baseline.

For EfficientNet, dropout and drop_connect_rate each mildly reduce the overfitting gap and slightly raise average validation F1 (best-validation F1 stays within baseline noise), while augmentation more substantially reduces the gap and raises average validation F1 at some cost to best-validation F1; all three are worth carrying into hyperparameter tuning. For ViT, both dropout and augmentation reduce the overfitting gap but also reduce average and best validation F1, suggesting neither is well-suited to ViT in isolation at the tested strength; we still test both jointly with a retuned

	Avg-train-mF1	Avg-val-mF1	Gap	Best-val-mF1
EfficientNet				
Baseline	0.8631	0.5880	0.2751	0.6874
+Dropout	0.8539	0.5966	0.2573	0.6822
+Drop_conn	0.8452	0.6038	0.2414	0.6933
+Augment	0.7549	0.5954	0.1595	0.6589
ViT				
Baseline	0.9596	0.7495	0.2101	0.7707
+Dropout	0.8830	0.7254	0.1576	0.7582
+Augment	0.8966	0.7203	0.1763	0.7672

Table 2: Regularization effects on overfitting mitigation.

Model	Loss	Test-mF1	Test bal-acc
EfficientNet	Weighted	0.6546	0.7036
	Unweighted	0.6405	0.6559
ViT	Weighted	0.7478	0.7511
	Unweighted	0.7159	0.7013

Table 3: Weighted vs. unweighted loss, best-tuned configuration, held-out test set. Accuracy columns report balanced accuracy (macro-recall); full validation/test breakdown in Appendix Table 9.

learning rate, batch size, and weight decay, since their interaction may differ from the untuned setting.

This reveals an architecture-dependent asymmetry: the same techniques that reliably help EfficientNet narrow ViT’s gap only by suppressing performance, not improving generalization. This motivates a combined, per-architecture hyperparameter search rather than a single recipe applied uniformly.

3.1.3 Hyperparameter Tuning and Loss Function Selection

Building on the individual regularization effects observed in Table 2, we jointly tune dropout, drop_connect_rate (EfficientNet only), augmentation, Phase 1/2 learning rates, weight decay, batch size, and warmup_ratio to identify a final, best-performing configuration for each architecture (full search grid in Appendix A.2). In parallel, we compare class-weighted against unweighted cross-entropy loss, given HAM10000’s substantial class imbalance.

Table 3 compares class-weighted against unweighted cross-entropy loss at each architecture’s best-tuned configuration (full validation/test breakdown in Appendix A.3). Unweighted loss reaches a marginally higher average validation F1 for both architectures, but weighted loss yields the smaller train-validation gap and better addresses HAM10000’s class imbalance, so we select class-weighted cross-entropy as final for both architectures. On the held-out test set that choice also proves the better one, by

0.014 macro-F1 on EfficientNet and 0.032 on ViT, so unweighted’s validation-average edge did not translate to test performance.

3.1.4 Baseline vs. Final Model Performance

Figure 1 illustrates the effect of our complete tuning pipeline on training dynamics for both architectures, plotting train and validation macro-F1 across epochs for the untuned baseline (orange) against the final selected configuration (blue). For both EfficientNet and ViT, the final model shows a visibly narrower train-val gap than the baseline, with training and validation curves converging more closely rather than diverging sharply, reflecting the combined effect of regularization, retuned learning rate and weight decay, and loss function selection. The final model’s validation curve also sits consistently above the baseline’s throughout training, confirming that these changes improved generalization rather than merely reducing the gap through weaker training performance.

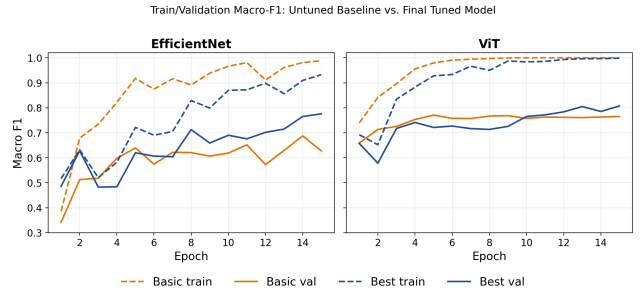


Figure 1: Learning curve of baseline and final tuned models for EfficientNet (left) and ViT (right)

3.2. Faithfulness

Figure 2 shows deletion and insertion curves, the predicted-class probability as pixels are removed (deletion) or added back (insertion) in the order each model’s heatmap ranks them, following RISE [9], against a baseline that ranks pixels at random. We summarize each curve by its area underneath, abbreviated AUC: a faithful heatmap makes deletion fall fast, so a lower deletion area is better, and makes insertion rise fast, so a higher insertion area is better. All curves are computed on the final tuned checkpoints selected in Section 3.1, the same checkpoints used for localization in Section 3.3, so all three axes are measured on identical models. Curves are macro-averaged over the seven predicted classes so that the majority class, which accounts for 60–69% of predictions depending on the model, cannot dominate; reported AUCs are taken on those macro-averaged curves.

Both explanations beat their own random-order baselines on both tests (Table 4). EfficientNet’s Grad-CAM is the

Table 4: Faithfulness AUCs on the final tuned checkpoints, macro-averaged over the seven predicted classes, $n = 1002$, seed 42.

Model (method)	Del. AUC ↓	Del. rand.	Ins. AUC ↑
EfficientNet-B0 (Grad-CAM)	0.284	0.528	0.736
ViT-B/16 (rollout)	0.661	0.791	0.582

Table 5: Faithfulness by predicted class, sorted by EfficientNet deletion AUC. n columns give the number of images predicted as that class by each model.

Class	n (EN/ViT)	EfficientNet		ViT	
		Del. ↓	Ins. ↑	Del. ↓	Ins. ↑
akiec	20/22	0.134	0.521	0.700	0.288
bcc	52/55	0.136	0.721	0.628	0.390
mel	203/88	0.151	0.735	0.345	0.485
df	19/15	0.187	0.753	0.573	0.468
bkl	84/115	0.309	0.570	0.613	0.856
vasc	24/16	0.488	0.971	0.805	0.783
nv	600/691	0.582	0.879	0.965	0.803

more faithful of the two: deletion AUC 0.284 against a random baseline of 0.528, and insertion AUC 0.736 against 0.283. ViT’s attention rollout is faithful by a narrower margin: deletion 0.661 against 0.791, insertion 0.582 against 0.387. Paired Wilcoxon signed-rank tests over the 1,002 test images confirm all four gaps ($p < 10^{-16}$ in every case) and confirm EfficientNet’s advantage over ViT on both deletion and insertion ($p = 1.1 \times 10^{-11}$ or smaller). The effect sizes differ sharply, however. EfficientNet’s ranking beats its random control on 79.2% of images for deletion and 89.0% for insertion, whereas ViT’s rollout beats random on only 55.3% of images for deletion. Rollout’s deletion advantage is statistically reliable at this sample size but small in practice, and we report it as such.

An earlier version of this evaluation, run on the pre-tuning checkpoints described in Section 2, gave EfficientNet an insertion AUC of 0.203 against a random baseline of 0.291, that is, worse than random ordering. That result does not survive on the tuned checkpoints, where the identical measurement, with the same mean-fill insertion canvas, gives 0.736 against 0.283. As with the localization results in Section 3.3, the sub-random insertion behaviour was a property of the unregularized configuration rather than of Grad-CAM or of the perturbation baseline itself.

Table 5 breaks the result down by predicted class and shows that faithfulness is far from uniform. EfficientNet’s deletion AUC is 0.134–0.151 on *akiec*, *bcc* and *mel*, but 0.582 on *nv*; ViT’s rollout reaches 0.965 on *nv*, essentially indicating that deleting its highest-ranked pixels barely changes the prediction. Both models therefore explain themselves least faithfully on the benign majority class and

most faithfully on the malignant and pre-malignant classes that carry the clinical cost of a miss. We note this pattern is best supported by the *mel* row ($n = 203$ and 88); the *akiec*, *df* and *vasc* rows rest on fewer than 25 images each and we report them without drawing individual conclusions.

3.3. Localization

Table 6 reports IoU and pointing-game hit-rate against HAM10000’s ground-truth lesion segmentation masks, computed with a shared, architecture-agnostic evaluator (percentile thresholding, top 20% of heatmap values) applied unchanged to both Grad-CAM and attention-rollout outputs, over all 1,002 test images (single seed, 42).

Table 6: Localization against ground-truth lesion masks, best checkpoints, $n = 1002$, seed 42.

Model	Method	IoU	PG
EfficientNet-B0	Grad-CAM	0.466±0.177	0.983
ViT-B/16	Rollout	0.401±0.186	0.650

EfficientNet’s Grad-CAM localizes more tightly than ViT’s rollout on both metrics. Table 7 compares localization before and after hyperparameter tuning. The tuned ViT configuration enables dropout, which the pretrained checkpoint ships at 0.0 (Section 2), but it also retunes learning rate, weight decay and batch size, so this is a comparison of configurations rather than a single-variable ablation.

Table 7: Localization before (Pre) vs. after (Best) hyperparameter tuning. The tuned ViT configuration enables dropout, which the checkpoint defaults set to 0.0.

Model	Metric	Pre	Best	Δ
EfficientNet-B0	IoU	0.484±0.182	0.466±0.177	−0.018
	PG	0.973	0.983	+0.010
ViT-B/16	IoU	0.300±0.190	0.401±0.186	+0.101
	PG	0.454	0.650	+0.196

ViT’s localization improves substantially with the tuned configuration: IoU rises about a third in relative terms and pointing-game hit-rate rises nearly 20 points, while EfficientNet’s numbers move only within noise. With attention regularized, ViT’s rollout maps are sharper and less diffuse, so the configuration that improved its accuracy also improved the quality of its explanations. Isolating dropout from the other retuned settings would require a dedicated single-variable run, which we did not perform.

3.4. Summary of results

Classification favors ViT, while both faithfulness tests and both localization metrics favor EfficientNet. All three axes are measured on the same tuned checkpoints, and the

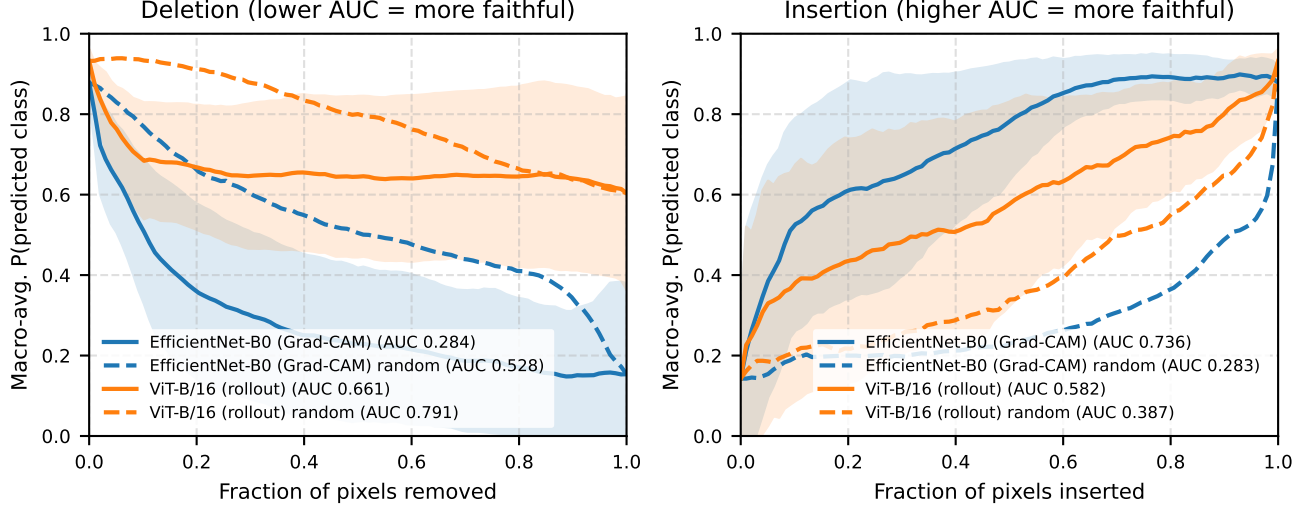


Figure 2: Deletion (left) and insertion (right) faithfulness curves on the final tuned checkpoints. Macro-averaged predicted-class probability vs. fraction of pixels removed/added, averaged over the seven predicted classes; shaded band is ± 1 std across per-class means. Dashed lines are each model’s random-ordering control. $n=1002$ test images, seed 42.

two explanation-quality axes agree: Grad-CAM is the more trustworthy explanation on this dataset.

4. Discussion

4.1. Problem Structure and Architectural Alignment

Skin lesion classification is a *bounded localization task*: diagnosis depends on fine-grained texture and morphology *within* the lesion, with robustness to background artifacts. This aligns naturally with EfficientNet’s local-to-global hierarchy and less obviously with ViT’s global-first attention. Our results support both perspectives. EfficientNet explains better on both faithfulness tests and on localization, consistent with locality-biased features tracking diagnostic texture directly, while ViT classifies better, consistent with global context such as asymmetry across the full lesion adding information a purely local model cannot see. We hold this loosely: the comparison rests on a single dataset and a single seed.

4.2. Model Components: Learned and Fixed

Learned: the 7-way classification head (trained from scratch) and the backbone (unfrozen and fine-tuned in Stage 2). **Fixed:** Grad-CAM and attention rollout, both post-hoc and parameter-free, plus softmax normalization and argmax classification. This split is also empirically supported: unfreezing the backbone already in Stage 1, then retuning its learning rate per variant, never exceeded the frozen-Stage-1 baseline’s validation macro-F1 of 0.7707 (Table 2) by more than epoch-to-epoch noise, so a frozen Stage 1 is at least as

good and cheaper to tune.

4.3. Input/Output Representations

Both networks take 224×224 RGB input normalized with ImageNet channel statistics, to stay aligned with each pretrained backbone. Output logits pass through softmax to probabilities, with the predicted class $\arg \max_i p_i$; the faithfulness tests (Section 3.2) track $p(\text{predicted class})$ directly, not the logits or heatmap.

4.4. Loss Function and Regularization

Class-weighted cross-entropy addresses the label imbalance (melanocytic nevus 67%, dermatofibroma 1%). Regularization behaved less predictably. Dropout alone left EfficientNet’s validation macro-F1 essentially flat across a 0.1–0.4 sweep (Section 3.1.2), which follows from where EfficientNet-B0 places its only dropout layer: immediately before the final linear layer, never touching the MB-Conv blocks that make up most of the unfrozen backbone. `drop_connect_rate` and weight decay each closed only a small part of the gap alone; augmentation, combined with moderate dropout and drop-connect rate, meaningfully narrowed it. ViT shows the opposite sensitivity: dropout 0.2 is its best single setting, while the same augmentation that helped EfficientNet reduces ViT’s validation macro-F1 at every dropout level tested. We read this not as ViT having “inherent attention-based regularization” but as having less inductive bias to fall back on, so perturbing input geometry interferes with learning positional structure rather than adding beneficial invariance as it does for a convolutional model.

4.5. Transfer Learning and Implementation

Starting from ImageNet-pretrained weights was necessary at this scale: both models converge within the 25-epoch schedule selected in Section 3.1, 10 epochs of Stage 1 and 15 of Stage 2, on 7,470 unique lesions, far fewer epochs and data than random initialization would need, protected by the two-stage schedule (Section 2). Both models are implemented in PyTorch with Hugging Face `transformers`, our own Grad-CAM [6] and attention-rollout [7] implementations, and RISE-style [9] faithfulness metrics; code and checkpoints are public (footnote, Section 2).

5. Future Work

Our central result, that the more accurate architecture is not the one with the more trustworthy explanations, opens two directions we think are worth pursuing beyond this study.

The first is training for faithfulness directly. We measured explanation quality only after training, as most work does, and found it can move opposite to accuracy. A natural question is whether faithfulness can instead be an objective: adding a term that rewards explanations whose highlighted pixels actually drive the prediction, and testing whether a model can be pushed toward both accurate and honest without sacrificing either. This would show whether the trade-off we observe is fundamental to these architectures or simply an artifact of optimizing for accuracy alone.

The second is establishing whether these explanation-quality gaps change clinical decisions. Our metrics score a heatmap against an expert mask, a proxy for whether a clinician would trust it, but not a measurement of trust itself. A study that puts CNN and transformer explanations in front of dermatologists and records how each affects diagnostic confidence and accuracy would tell the field whether the faithfulness and localization differences we quantify matter at the bedside, or whether accuracy is what a clinician ultimately relies on.

Limitations: all results are single-seed (42) on one dataset, so the gaps we report are not accompanied by variance estimates. We also compare each architecture with the attribution method standard for it, Grad-CAM against attention rollout, which confounds architecture with explanation method; an architecture-agnostic attributor applied to both would separate the two. Finally, EfficientNet-B0 (5.3M parameters) and ViT-B/16 (86M) differ substantially in capacity, so the accuracy gap is not attributable to inductive bias alone.

Broader impact: the architecture-specific overfitting pattern (Section 3.1.2) follows from where each architecture places its regularization, not from anything skin-specific, so we expect it and our faithfulness/localization methodology to generalize to other small medical imaging datasets;

HAM10000’s skew toward European-institution data remains a limitation for any clinical deployment.

6. Conclusion

We compared a convolutional network and a vision transformer on HAM10000 across classification accuracy, faithfulness, and localization. ViT-B/16 classifies better (0.748 vs. 0.655 test macro-F1), while EfficientNet’s Grad-CAM explains better on both faithfulness tests and on both localization metrics. The trade-off the Introduction set out to test therefore holds, on all three axes measured over the same tuned checkpoints: the architecture producing more trustworthy explanations is not the one that classifies most accurately.

A second finding came out of the hyperparameter work. The two architectures overfit for different reasons and need different regularization (Section 3.1.2), and in both cases the tuned configuration changed explanation quality, not just accuracy: on the ViT it improved localization (IoU +0.101, pointing-game +0.196, Section 3.3), while on EfficientNet it turned an insertion AUC that had fallen below its own random baseline (0.203 vs. 0.291) into one well above it (0.736 vs. 0.283, Section 3.2). Regularization and hyperparameter choices should therefore be judged on their effect on interpretability too.

The consequence for deployment is that the two questions must be asked separately. Selecting on validation accuracy, as the standard recipe does, would have chosen the model whose explanations we found less trustworthy, which no accuracy number would reveal.

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A. Appendix

A.1. Full Fine-Tuning Strategy Depth Sweep

Table 8 gives the complete depth sweep behind the summary in Table 1 (Section 3.1.1).

Strategy 1					Strategy 2				
Unfreeze Layer	2	4	6	all	Unfreeze Layer	2	4	6	
EfficientNet					EfficientNet				
Train-mF1	0.8540	0.9341	0.9571	0.9880	Train-mF1	0.9943	0.9919	0.9986	
Val-mF1	0.5631	0.5998	0.6131	0.6874	Val-mF1	0.6496	0.6488	0.6866	
ViT					ViT				
Train-mF1	0.8960	0.9802	0.9984	1	Train-mF1	1	1	1	
Val-mF1	0.6956	0.7211	0.7477	0.7707	Val-mF1	0.7501	0.7437	0.7577	

Table 8: Strategy 1 vs. Strategy 2 full depth sweep, training and validation MacroF1.

A.2. Full Hyperparameter Search Grid

Full grid swept jointly for each architecture in Section 3.1:

- dropout: {0.1, 0.2, 0.3, 0.4}
- drop_connect_rate: {0.1, 0.2, 0.3, 0.4}
- augmentation: {True, False}
- Phase 1 lr: $\{1e^{-4}, 4e^{-4}, 8e^{-4}, 1e^{-3}, 5e^{-3}, 8e^{-3}, 1e^{-2}\}$
- Phase 2 lr: $\{1e^{-5}, 2e^{-5}, 4e^{-5}, 8e^{-5}, 1e^{-4}, 5e^{-4}, 8e^{-4}\}$
- weight decay: {0.01, 0.05, 0.1, 0.2}
- batch size: {8, 16, 32}
- warmup_ratio: {0.1, 0.2}

A.3. Full Weighted vs. Unweighted Loss Comparison

Table 9 gives the full validation/test breakdown behind the summary in Table 3 (Section 3.1).

	Avg_tr_mF1	Avg_v_mF1	Gap	Best_tr_mF1	Best_v_mF1	Best_v_bal-acc	Test_mF1	Test_bal-acc
EfficientNet								
Weighted	0.6394	0.5601	0.0793	0.9323	0.7759	0.7964	0.6546	0.7036
Unweighted	0.7010	0.5956	0.1054	0.9853	0.7378	0.7428	0.6405	0.6559
ViT								
Weighted	0.7176	0.6069	0.1107	0.9982	0.8074	0.7893	0.7478	0.7511
Unweighted	0.7495	0.6145	0.1350	0.9999	0.7835	0.7684	0.7159	0.7013

Table 9: Weighted vs. unweighted loss function comparison, full breakdown. Accuracy columns report balanced accuracy (macro-recall). Best_tr_mF1/Best_v_mF1/Best_v_bal-acc are reported at the epoch of peak validation Macro-F1 (the checkpointed epoch); test metrics are from that checkpoint.

A.4. Model Architectures

EfficientNet-B0 — Fine-Tuning Architecture (HAM10000, 7-class skin lesion classification)

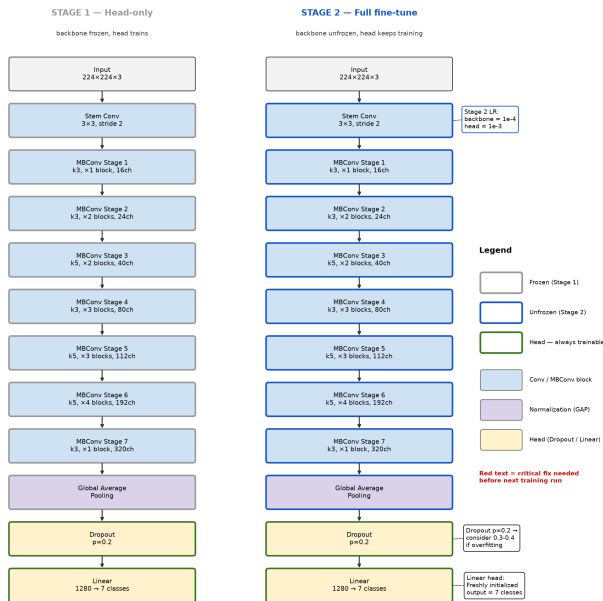


Figure 3: EfficientNet architecture.

VIT-B/16 — Fine-Tuning Architecture (HAM10000, 7-class skin lesion classification)

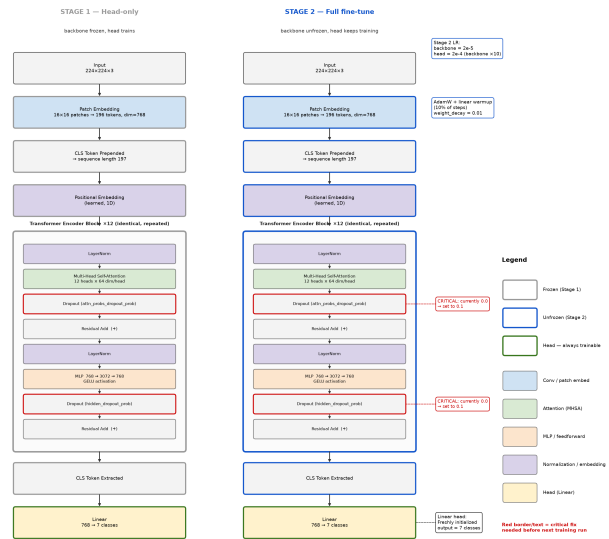


Figure 4: ViT-B/16 architecture.